The Steerable Bispectrum for Non-Abelian Groups

Bridging Equivariant Networks and Complete Invariant Theory

Johan Mathe

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Two worlds, one gap

Steerable CNNs

Weiler & Cesa 2019

- Equivariant by design
- Learned filters on intertwiner basis
- Invariance via norm / gated nonlinearities

→ Discards phase information

→ No reconstruction guarantees

G-Bispectrum

Kakarala 1992, Mataigne et al. 2024

- Complete invariant (up to group action, under non-degeneracy)
- Signal recoverable from bispectrum (generically)
- Selective: $O(|G|)$ instead of $O(|G|^2)$ coefficients

→ Requires proper Fourier coefficients

→ Not embedded in learned architectures

Gap: What seems to be missing is a principled identification between steerable feature coefficients and the Fourier objects required by the non-abelian bispectrum.

What does "steerable" mean?

A basis of filter functions is **steerable** if group actions reduce to linear operations on the coefficients:

$$\mathbf{Y}(g \cdot x) = \rho(g)\,\mathbf{Y}(x)$$

where $\mathbf{Y}$ is vector-valued and $\rho(g)$ is a matrix representation.

Abelian ($\mathrm{SO}(2)$, C_n):
Each basis element Y_l transforms by a *scalar* character: $Y_l(g \cdot x) = \chi_l(g)\, Y_l(x)$.
E.g. $e^{ik\theta} \mapsto e^{ik\phi} \cdot e^{ik\theta}$

Non-abelian (D_n, O):
Basis elements within an isotypic block *mix* under the group action via a matrix $\rho(g)$.
E.g. the 2D irrep E of D_4

Analogy: a spotlight on a gimbal. You turn a dial (apply $\rho(g)$) and the beam rotates analytically — no grid resampling.

Reference: Freeman & Adelson, *The Design and Use of Steerable Filters*, IEEE TPAMI 1991.

What is the bispectrum?

Abelian case (C_n, $\mathrm{SO}(2)$): scalar triple product

$$\beta_{k_1,k_2} = \hat{f}_{k_1} \cdot \hat{f}_{k_2} \cdot \hat{f}^*_{k_1+k_2}$$

Non-abelian case (D_n, O, $\mathrm{SO}(3)$): schematic CG-based formula

$$\beta_{\rho_1,\rho_2} = \left[\mathcal{F}_{\rho_1} \otimes \mathcal{F}_{\rho_2}\right] C_{\rho_1,\rho_2} \left[\bigoplus_{\rho \in \rho_1 \otimes \rho_2} \mathcal{F}^\dagger_\rho\right] C^\dagger_{\rho_1,\rho_2}$$

where $\mathcal{F}_\rho$ denotes the matrix-valued Fourier coefficient at irrep ρ, and C_{ρ_1,ρ_2} is the CG change-of-basis for $\rho_1 \otimes \rho_2$.

Key property: under standard non-degeneracy assumptions (invertibility of relevant Fourier blocks), the bispectrum is a *complete* invariant — it recovers f up to group action. The power spectrum $|\hat{f}_k|^2$ is not (loses phase).

Selective bispectrum (Mataigne et al. 2024): only $O(|G|)$ coefficients needed instead of $O(|G|^2)$, preserving completeness.

Timeline: two parallel lines of work

Bispectrum theory

- 1992 — Kakarala: triple correlation on groups
- 2012 — Kakarala: complete invariant, SO(3) recovery
- 2024 — Mataigne et al.: selective G-bispectrum

Steerable / equivariant CNNs

- 2017 — Worrall: H-Nets (circular harmonics)
- 2019 — Weiler & Cesa: general E(2)-steerable
- 2020 — Oreiller et al.: SO(2) bispectrum CNN

This work

Part I: The Abelian Case

How the steerable bispectrum works for $\mathrm{SO}(2)$

$\mathrm{SO}(2)$ steerable bispectrum

Filter: $\Psi_k(r,\theta) = w_k(r) \cdot e^{ik\theta}$ (learned radial profile $\times$ circular harmonic)

Convolution: produces per-pixel steerable coefficients $\hat{\Theta}_k$

Steerability: rotate input by $\phi \Rightarrow \hat{\Theta}_k \mapsto e^{ik\phi}\hat{\Theta}_k$

Bispectrum: triple product cancels the phase

$$\beta_{k_1,k_2} = \hat{\Theta}_{k_1} \cdot \hat{\Theta}_{k_2} \cdot \hat{\Theta}^*_{k_1+k_2}$$

since $e^{ik_1\phi} \cdot e^{ik_2\phi} \cdot e^{-i(k_1+k_2)\phi} = 1$.

This is the same scalar triple-product mechanism as `CnonCn`, applied per-pixel to circular-harmonic responses.

Why it works: $\mathrm{SO}(2)$ is abelian $\Rightarrow$ all irreps are 1D characters $\Rightarrow$ scalar steerable coordinates align with scalar Fourier modes.

The existing implementation (Oreiller et al. 2022)

The BCHConv2D layer in voreille/2d_bispectrum_cnn:

```
def call(self, inputs):
    x = self.conv_ch(inputs)     # circular harmonic convolution
    outputs = list()
    for n1, n2 in self.indices:
        bispectrum = (x[..., n1] * x[..., n2] *
                      tf.math.conj(x[..., n1 + n2]))
        outputs.append(tf.math.real(bispectrum))
        if n1 != 0:
            outputs.append(tf.math.imag(bispectrum))
    x = tf.concat(outputs, axis=-1)
```

- TensorFlow/Keras, not PyTorch
- Redundant index set ($\sim n^2/2$ pairs via symmetry reduction)
- No selectivity, no inversion guarantees
- Learned discrete radial profiles (pixel masks), not Fourier-Bessel basis

What's missing for the $\mathrm{SO}(2)$ case

	Oreiller et al.	Our library	Potential
Bispectral pairs	$\sim n^2/2$ (symmetry)	n (selective)	Same accuracy, fewer channels
Inversion	None	✓ (bootstrap)	Reconstruct from invariants
Radial basis	Learned pixel masks	Bessel $J_n(\lambda_{nk} r)$	Orthogonal, complete
Framework	TensorFlow	PyTorch	GPU-optimized

At $n_{\text{harmonics}} = 4$: selective gives 4 vs their 6 coefficients — negligible.

At $n_{\text{harmonics}} = 32$: selective gives 32 vs $\sim$528 coefficients.

At $n_{\text{harmonics}} = 64$: selective gives 64 vs $\sim$2080 coefficients.

Selectivity matters when angular resolution is high, and becomes critical in 3D where full bispectral constructions scale poorly with harmonic bandwidth.

Why the abelian case is easy

For an abelian group G, all irreps are 1D characters $\chi_k : G \to \mathbb{C}$.

The intertwiner space $\mathrm{Hom}^G(\pi, \chi_k)$ has dimension equal to the multiplicity of χ_k in π.

For the regular representation on $\mathbb{Z}/n\mathbb{Z}$: every character appears exactly once.

In the regular abelian case, the equivariant decomposition coincides with the Fourier decomposition: every irrep is 1D, appears once, and $L = M = n$.

Hence the steerable coefficients $\hat{\Theta}_l$ can be identified with the Fourier coefficients $\hat{f}_l$.

$\Rightarrow$ The standard bispectrum formula applies directly.

For non-abelian groups, the simple scalar identification between steerable coefficients and Fourier coefficients breaks.

Part II: The Non-Abelian Problem

Why $L \neq M$ blocks the standard approach

The $L \neq M$ problem

Consider $H = D_4$ with a **trivial output representation** ρ_0.

A trivial-output head accesses only the A_1-isotypic component of the input fiber π.
But D_4 has $M = 5$ irreps: A_1, A_2, B_1, B_2, E (dims $1, 1, 1, 1, 2$).

Count-of-types intuition (not a dimensional comparison):

	Irreps accessed by trivial output	Total irreps of H
$\mathrm{SO}(2)$, C_n (abelian)	all n (each 1D)	n
D_4, $\rho = \rho_0$	A_1 only	5 (incl. 2D E)
D_8, $\rho = \rho_0$	A_1 only	7 (incl. 2D irreps)

The steerable conv gives access only to the A_1-isotypic coordinates, while the bispectrum requires Fourier data from all 5 irreps of D_4, including matrix-valued blocks.

Why $L \neq M$

The number of intertwiner basis functions is:

$$L = \dim \mathrm{Hom}^H(\pi, \rho) = \sum_{\text{irreps } \psi} m_\psi^\pi \cdot m_\psi^\rho$$

For $\rho = \rho_0$ (trivial): $m_\psi^{\rho_0} = 0$ for all $\psi \neq \rho_0$, so $L = m_{\rho_0}^\pi$.

This counts only how many times the *trivial* irrep appears in π — it says nothing about the other irreps.

The Fourier coefficients $\mathcal{F}(f)_\psi$ for the non-trivial irreps (A_2, B_1, B_2, E) are **invisible** to a steerable conv with trivial output.

For abelian groups in the regular representation: every irrep is 1D and appears with multiplicity one, so the equivariant decomposition aligns with the Fourier decomposition and $L = M$.

What the coefficients actually are

The steerable conv produces per-pixel coefficients $\hat{\Theta}_l$, which are projections onto the ρ_0-**isotypic component** of the input fiber π:

- **Visible** through a trivial-output head: the A_1-isotypic component, of multiplicity $m^{\pi}_{A_1} = 3$
- Not visible through a trivial-output head: the A_2, B_1, B_2, E isotypic components (4 irreps, including the matrix-valued E block)

$\Rightarrow$ The bispectrum computed from $\hat{\Theta}_l$ alone is *incomplete*.

$\Rightarrow$ For a complete invariant, we need Fourier data for all irreps $\mathcal{F}_{\psi}$.

Information flow: the problem and our solution

Signal f → Steerable conv (reg. rep output) → $|H|$ coefficients in $\bigoplus \psi \otimes \psi^*$ → Recover $\hat{f}(\psi)$ via A_ψ (P4) → G-Bispectrum (invariant)

Group	Solution	Status
$\mathrm{SO}(2)$, C_n (abelian)	Identity ($L = M$)	✓ Solved (Oreiller 2020)
D_n (non-abelian)	Regular-rep fiber + P4	Proved for model case (this work)
O (octahedral)	Same rep-theoretic construction	Expected to apply

Previously, steerable CNNs for D_n were stuck with **norm** (loses phase) or **gated** (ad-hoc). P4 with the regular-rep fiber gives a principled construction and a clear sufficient condition for bispectral pooling.

Part III: The Solution

Regular-representation output fiber and Proposition P4

Construction: regular-representation output fiber

Key idea: design the output fiber to be the **regular representation** of H.

For each irrep ψ, the Fourier block $\hat{f}(\psi) \in \psi \otimes \psi^*$ has dimension d_ψ^2. To recover it, the ψ-typed output needs multiplicity d_ψ (not just 1).

For D_4, the regular representation $\mathbb{C}[D_4] \cong \bigoplus_\psi \psi \otimes \psi^*$ gives:

Irrep ψ	d_ψ	Canonical mult. (reg. rep)	Output $\psi \otimes \psi^*$	Scalar dim	Fourier block
A_1	1	1	$A_1 \otimes A_1^*$	1	$\hat{f}(A_1) \in \mathbb{C}$
A_2	1	1	$A_2 \otimes A_2^*$	1	$\hat{f}(A_2) \in \mathbb{C}$
B_1	1	1	$B_1 \otimes B_1^*$	1	$\hat{f}(B_1) \in \mathbb{C}$
B_2	1	1	$B_2 \otimes B_2^*$	1	$\hat{f}(B_2) \in \mathbb{C}$
E	2	2	$E \otimes E^*$	4	$\hat{f}(E) \in \mathbb{C}^{2\times 2}$
Total				$8 = \lvert D_4 \rvert$	All 5 Fourier blocks

Total dimension: $1 + 1 + 1 + 1 + 4 = 8 = |D_4|$, recovering all 5 Fourier blocks.

Why multiplicity d_ψ matters

Key: a single E-field (2 scalars) cannot recover the E-Fourier block (2×2 matrix, 4 scalars).

You need **two copies** of E, i.e. $E \otimes E^* \cong \mathrm{End}(E)$.

More generally, $\hat{f}(\psi) \in \psi \otimes \psi^*$ has d_ψ^2 scalar entries. A ψ-typed output with multiplicity $m < d_\psi$ produces only $m \cdot d_\psi$ scalars — not enough to determine the full Fourier block.

The regular representation is the *canonical* choice: it gives each irrep ψ exactly multiplicity d_ψ, which is both necessary and sufficient.

Proposition P4 (proved for $V = \mathbb{C}[H]$)

Proposition. Let $V = \mathbb{C}[H]$ (regular representation). Let $T_\psi \in \mathrm{Hom}^H(V, \psi \otimes K)$ be an equivariant map with output type $\psi \otimes K$. Then there exists a linear map $A_\psi : \psi^* \to K$ such that

$$T_\psi(f) = (I_\psi \otimes A_\psi)\, \hat{f}(\psi).$$

If $K \cong \psi^*$ and A_ψ is an isomorphism, then $\hat{f}(\psi)$ is linearly recoverable.

Consequence: with a regular-rep output fiber ($K = \psi^*$ for each irrep), the steerable conv output is linearly equivalent to the full set of Fourier blocks. Selectivity (Mataigne et al. 2024) then applies directly.

Proof of P4 (step 1: decomposition)

Step 1 (Peter–Weyl). The regular representation decomposes as

$$V = \mathbb{C}[H] \;\cong\; \bigoplus_{\psi \in \widehat{H}} \psi \otimes \psi^*,$$

where each summand $V_\psi = \psi \otimes \psi^*$ is the ψ-*isotypic component*, a $d_\psi \times d_\psi$ matrix space. The group acts on the left factor: $h \cdot (u \otimes \lambda) = (\psi(h)u) \otimes \lambda$.

The orthogonal projection $P_\psi : V \to V_\psi$ is exactly the Fourier transform at ψ:

$$P_\psi f \;\longleftrightarrow\; \hat{f}(\psi) = \sum_{g \in H} f(g)\, \psi(g)^* \;\in\; \mathbb{C}^{d_\psi \times d_\psi}.$$

Write $V = V_\psi \oplus V_\psi^\perp$, where $V_\psi^\perp = \bigoplus_{\phi \neq \psi} V_\phi$.

Proof of P4 (step 2: Schur kills wrong frequencies)

Step 2 (Schur's lemma). The map $T_\psi : V \to \psi \otimes K$ is H-equivariant, with the target transforming as ψ in the first factor.

Restrict T_ψ to a summand $V_\phi = \phi \otimes \phi^*$ with $\phi \neq \psi$. This restriction is an equivariant map $\phi \otimes \phi^* \to \psi \otimes K$. Since H acts on the first factors, this requires an equivariant map $\phi \to \psi$.

Schur's lemma: if $\phi \not\cong \psi$, every equivariant map $\phi \to \psi$ is **zero**.

Therefore T_ψ annihilates $V_\psi^\perp$ and factors through the Fourier projection:

$$T_\psi = \tilde{T}_\psi \circ P_\psi, \qquad \tilde{T}_\psi : \underbrace{\psi \otimes \psi^*}_{=V_\psi} \longrightarrow \psi \otimes K.$$

Consequence: equivariance forces frequency selectivity — T_ψ can only access the Fourier block $\hat{f}(\psi)$.

Proof of P4 (step 3: structure on the right frequency)

Step 3 (classification of equivariants). We must classify all H-equivariant maps

$$\tilde{T}_\psi : \psi \otimes \psi^* \longrightarrow \psi \otimes K,$$

where H acts on ψ (first factor) and acts trivially on ψ^* and K.

By Schur's lemma applied to the ψ-factor: any equivariant map $\psi \to \psi$ is a scalar multiple of the identity. Therefore $\tilde{T}_\psi$ must act as the identity on the first factor:

$$\tilde{T}_\psi = I_\psi \otimes A_\psi, \qquad A_\psi : \psi^* \to K \text{ (arbitrary linear map).}$$

Combining steps 1–3:

$$\boxed{T_\psi(f) = (I_\psi \otimes A_\psi)\, \hat{f}(\psi).}$$

When $K \cong \psi^*$ and A_ψ is invertible, we recover $\hat{f}(\psi) = (I_\psi \otimes A_\psi^{-1})\, T_\psi(f)$. $\square$

Remark: the proof uses only Peter–Weyl and Schur's lemma — it holds for *any* finite group H, not just D_4.

Approach 2: direct triple correlation

Idea: bypass the steerable-to-Fourier identification entirely. Define the triple correlation directly on vector-valued signals with an induced representation action.

What we have proven (in our notes):

1. A triple correlation $T(f)_{g_1,g_2}$ for vector-valued f with action $T_{tr}[f](x) = \rho_H(r)\, f\big((tr)^{-1}x\big)$.
2. **Invariance**: $T(T_h[f]) = T(f)$ for all $h \in H$ (change of variable $r' = rr_h$, $t' = R_h t + t_h$).
3. **Bispectrum recovery**: FT of $T(f)$ recovers the standard CG-based bispectrum formula.

This confirms the bispectrum is well-defined for induced-representation signals. In the regular-rep model case, P4 resolves the steerable-to-Fourier identification. For `escnn`-style induced features over $\mathbb{R}^2$, the identification of the intertwiner basis with the Fourier multiplicity space remains to be verified.

Proof status from our derivations

Result	Status	Where
TC definition for induced rep	✓Done	Eq. in our notes
TC invariance proof	✓Done	Change of variable $r' = rr_h$, $t' = Rt_h + t$
Bispectrum = FT of TC	✓Done	CG-based derivation
Standard CG formula recovered	✓Done	$\beta = [\mathcal{F}_{\rho_1} \otimes \mathcal{F}_{\rho_2}]C[\bigoplus \mathcal{F}_\rho^\dagger]C^\dagger$
$\mathcal{F}_\rho$ from steerable conv (P4)	✓Done	Schur + Peter–Weyl on regular rep
Selectivity transfers	✓Done	Direct from P4 + Mataigne et al.
Channel count: $\lvert H\rvert$ scalars suffice	✓Done	Regular rep has dim $\lvert H\rvert$
`escnn` fiber = regular rep	Open	Implementation question

Remaining question: the theorem holds for the regular representation. In `escnn`, the fiber is an architecture choice. The open task is to verify that `escnn` allows setting the output type to $A_1 \oplus A_2 \oplus B_1 \oplus B_2 \oplus 2E$ and to compute A_ψ from the intertwiner basis.

Selectivity transfers (from P4)

Since P4 establishes that the steerable conv with regular-rep fiber recovers all Fourier blocks, Mataigne et al. (2024) selectivity applies directly:

Channel count per pixel, for group H:

Invariant	**Channels**	**Complete?**†	**Invertible?**†
Norm (power spectrum)	$O(\|H\|)$	No	No
Full bispectrum	$O(\|H\|^2)$	Yes	Yes
Selective bispectrum	$\mathbf{O(\|H\|)}$	**Yes**	**Yes**

† Under standard non-degeneracy assumptions (Kakarala 1992, Mataigne et al. 2024).

Concrete numbers for D_n:

Group	$\|H\|$	Full bisp. channels	Selective channels
D_4	8	64	8
D_8	16	256	16
D_{16}	32	1024	32

Selective bispectrum is the **only** option that is both generically complete and tractable.

Proof summary

ID	Statement	Depends on	Status
P4	$T_\psi(f) = (I_\psi \otimes A_\psi)\hat{f}(\psi)$ for any finite H	—	✓ Done
P5	Bispectrum from regular-rep steerable conv	P4	✓ Done
P6	Selectivity transfers to steerable setting	P4, P5	✓ Done
P7	Channel count: $\lvert H \rvert$ scalars for complete invariant	P6	✓ Done
E1	`escnn` output type = regular rep of D_4	—	Open
E2	Compute A_ψ from `escnn` intertwiner basis	E1	Open

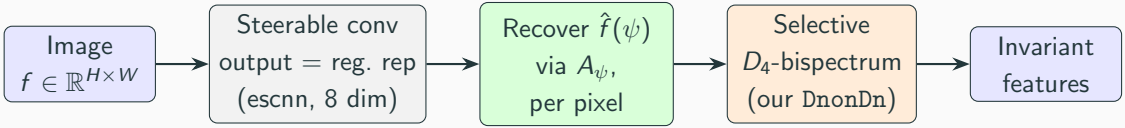
Theory is complete for the regular-representation model case.

Remaining work: realize this construction faithfully in `escnn` field-type language.

Part IV: Directions and Experiments

What to build and how to test it

Architecture: D_4 steerable bispectrum layer



- **Steerable conv**: `escnn` with output type $A_1 \oplus A_2 \oplus B_1 \oplus B_2 \oplus 2E$ (regular-rep fiber, 8 scalars/pixel)
- **Fourier extraction**: invert A_ψ to recover $\hat{f}(\psi)$ from each isotypic block (P4)
- **Selective bispectrum**: use our `DnonDn` module per pixel
- **Output**: D_4-invariant feature maps, ready for downstream task

Drop-in replacement for norm/gated nonlinearities in equivariant architectures.

Experiment 1: rotated CIFAR-10/100

Setup: Standard benchmark for rotation-invariant classification.

Method	Invariance	Complete?†	Source
Standard CNN + augmentation	None (learned)	N/A	Baseline
Steerable CNN + norm	D_4	No	Weiler & Cesa 2019
Steerable CNN + $\mathrm{SO}(2)$ bisp.	$\mathrm{SO}(2)$	Yes (rot)	Oreiller 2020
Steerable CNN + D_4 sel. bisp.	D_4	**Yes (rot+ref)**	**Ours**

† Generically, under non-degeneracy of relevant Fourier blocks.

Expected: D_4 selective bispectrum yields a generically complete invariant for rotations and reflections; norm is D_4-invariant too, but incomplete. Selective version matches full bispectrum with $8\times$ fewer channels.

Why impactful: First head-to-head comparison of bispectrum vs norm/gate nonlinearities in steerable CNNs.

Experiment 2: medical image segmentation

Setup: Reproduce Oreiller et al. (MIDL 2022) on MoNuSeg (nucleus segmentation in histopathology). Replace their $\mathrm{SO}(2)$ bispectral U-Net with our D_4 selective bispectral U-Net.

Why D_4 matters here:

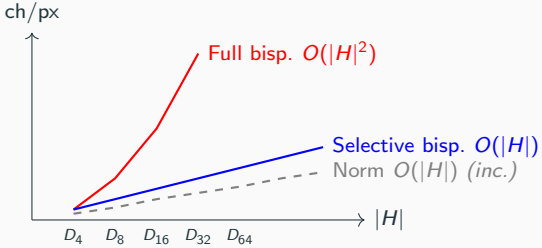
- Cell nuclei have no preferred chirality $\Rightarrow$ reflection invariance is natural
- $\mathrm{SO}(2)$ bispectrum handles rotations only
- D_4 bispectrum handles rotations *and* reflections

Expected: Better rotation+reflection robustness on their benchmark, same architecture.

Why impactful: Direct improvement on published results, on their own data, with a principled mathematical upgrade.

Experiment 3: scaling analysis

How does the invariant feature dimension scale with group size?



Selective bispectrum is the only option that is both:

- **Generically complete** (unlike norm, which loses phase)
- **Tractable** (unlike full bispectrum, which blows up quadratically)

Selectivity becomes increasingly important in 3D, where full bispectral constructions scale poorly with harmonic bandwidth.

Takeaway

1. **Abelian steerable bispectrum** ($\mathrm{SO}(2)$) works: same scalar triple-product as `CnonCn`, applied per-pixel to circular-harmonic responses.
2. For **non-abelian groups** (D_n, O), a single output type does not expose Fourier data for all irreps. Fix: **regular-representation output fiber**.
3. **Proposition P4** (proved for $V = \mathbb{C}[H]$): $T_\psi(f) = (I_\psi \otimes A_\psi)\hat{f}(\psi)$. Multiplicity d_ψ in output needed for d_ψ-dim irreps. Selectivity transfers.
4. **Design**: for D_4, output $= A_1 \oplus A_2 \oplus B_1 \oplus B_2 \oplus 2E$ (8 scalars $= |D_4|$).
5. **Target**: NeurIPS 2026. Theory + `SteerableBispectrum` layer + experiments on rotated CIFAR and MoNuSeg.

Remaining work: map this construction to `escnn` — verify the fiber can be set to the regular rep and compute A_ψ from the intertwiner basis.